

Analog IC Design Implementation and Verification

Current Mirrors and Amplifiers – Weeks 3 and 4 Labs

**Course Version IC 23.1 ISR11.29 with SPECTRE 24.1 ISR1
Lab Manual**

Revision 2.0

© 1990-2025 Cadence Design Systems, Inc. All rights reserved.

Printed in the United States of America.

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

Trademarks: Trademarks and service marks of Cadence Design Systems, Inc. (Cadence) contained in this document are attributed to Cadence with the appropriate symbol. For queries regarding Cadence trademarks, contact the corporate legal department at the address shown above or call 1-800-862-4522.

All other trademarks are the property of their respective holders.

Restricted Print Permission: This publication is protected by copyright, and any unauthorized use of this publication may violate copyright, trademark, and other laws. Except as specified in this permission statement, this publication may not be copied, reproduced, modified, published, uploaded, posted, transmitted, or distributed in any way, without prior written permission from Cadence. This statement grants you permission to print one (1) hard copy of this publication subject to the following conditions:

- The publication may be used solely for personal, informational, and noncommercial purposes;

- The publication may not be modified in any way;

- Any copy of the publication or portion thereof must include all original copyright, trademark, and other proprietary notices, and this permission statement and

- Cadence reserves the right to revoke this authorization at any time, and any such use shall be discontinued immediately upon written notice from Cadence.

Disclaimer: Information in this publication is subject to change without notice and does not represent a commitment on the part of Cadence. The information contained herein is the proprietary and confidential information of Cadence or its licensors and is supplied subject to and may be used only by Cadence customers in accordance with a written agreement between Cadence and the customer.

Except as may be explicitly set forth in such agreement, Cadence does not make and expressly disclaims any representations or warranties as to the completeness, accuracy, or usefulness of the information contained in this document. Cadence does not warrant that the use of such information will not infringe any third party rights, nor does Cadence assume any liability for damages or costs of any kind that may result from the use of such information.

Restricted Rights: Use, duplication, or disclosure by the Government is subject to restrictions as set forth in FAR52.227-14 and DFAR252.227-7013 et seq. or its successor.

Table of Contents

Analog IC Design Implementation and Verification *Current Mirrors and Amplifiers*

Module 1:	DC Analysis of Current Mirror Circuit.....	4
Lab 1-1	Design of Current Mirror Circuit.....	5
	Creating a Library and Cellview	5
	Invoking the Add Instance Form in the Cellview	7
	Adding Components to a Schematic	9
	Adding Ground and VDD to the Schematic.....	13
	Creating Wires to a Schematic.....	14
	Saving a Design	15
	Netlisting the Design.....	16
Lab 1-2	Simulation Using ADE Explorer and Virtuoso VA.....	19
	Starting the Virtuoso ADE Explorer and Creating a <i>maestro</i> View.....	19
	Setting the Model Libraries.....	20
	Setting Design Variables.....	21
	Choosing Analyses.....	23
	Saving and Plotting Simulation Outputs	25
	Running the Simulation	26
	DC Parametric Analysis.....	28
	Simulating the Parametric Analysis	29

Module 1: DC Analysis of Current Mirror Circuit

Lab 1-1 Design of Current Mirror Circuit

Objective: To use the basic schematic capture commands to build a current mirror circuit.

In this lab, you open the schematic capture environment window and add components, wires, and pins. The schematic is then checked and saved.

Creating a Library and Cellview

1. Open the Terminal window, open the directory of **basic_analog_labs**, by entering this command in the terminal window:

```
cd ~/basic_analog_labs
```

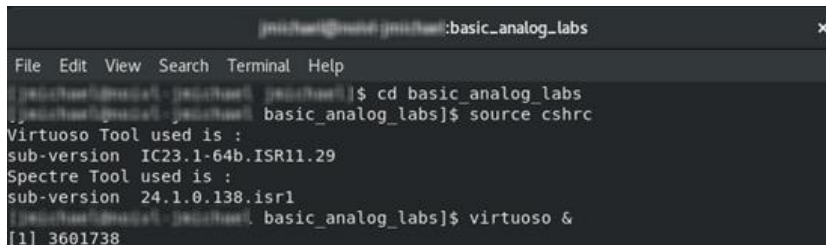
2. In that directory, source the License file by entering this command in the terminal window:

```
source cshrc
```

3. Type the following command to open CIW:

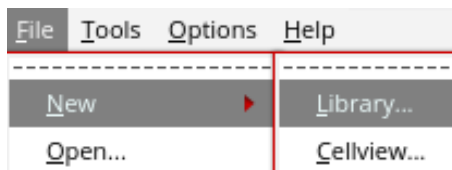
```
virtuoso &
```

The following screenshot gives the view of the terminal window on these commands executed:

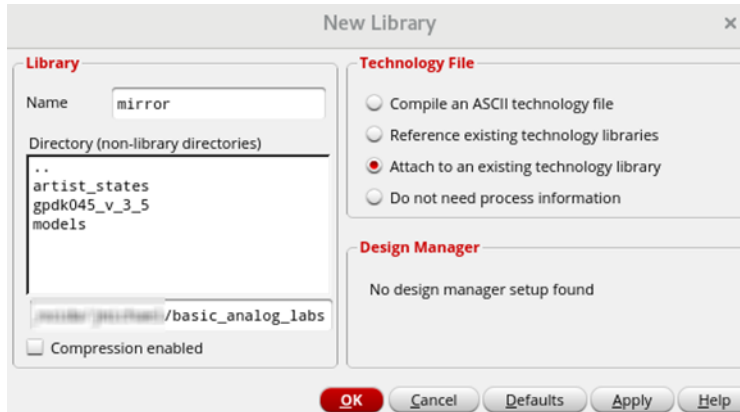


```
prashant@prashant:~/basic_analog_labs$ cd basic_analog_labs
prashant@prashant:~/basic_analog_labs$ source cshrc
Virtuoso Tool used is :
sub-version IC23.1-64b.ISR11.29
Spectre Tool used is :
sub-version 24.1.0.138.isr1
prashant@prashant:~/basic_analog_labs$ virtuoso &
[1] 3601738
```

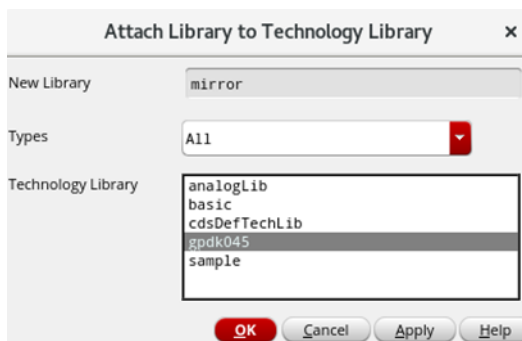
4. From CIW, choose **File – New – Library**.



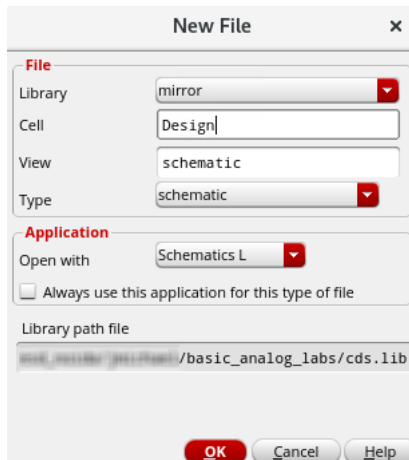
5. Name the Library as **mirror** and select the radio button for **Attach to an existing technology Library** and click on **OK**.



6. In the Attach Library to Technology Library form, select on the **gpd045** library and click on **OK**.



7. Create a new cell view under the same **mirror** library by choosing **File – New – Cellview** created, name it as **Design**, and click on **OK** to open the schematic view.



Invoking the Add Instance Form in the Cellview

The instructions in this section describe how to build the current mirror circuit schematic shown in the following snapshot.

1. With your cursor in the schematic window, go to the top of the schematic window to the toolbar. Click the **Create Instance** icon to display the Add Instance form. 



Note: You can also choose **Create – Instance** from the pull-down menus or use the **I** key to pop up the form.

2. In the Add Instance form, click on **Browse** to open the Library Browser.

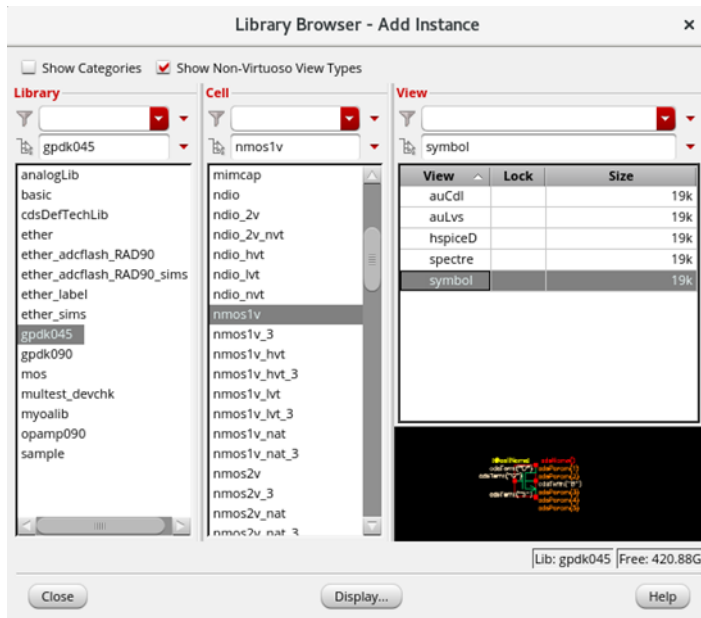
3. In the **Library Browser – Add Instance** form, select **gpd045** in the library column. The Cell column of this form now displays the available cells in that library.
4. Use the scroll bar to the right of the Cell column to locate **nmos1v** and **click** to select this cell. The Library Browser form now displays several views for the *nmos1v* cell.

Note: You can use the filtering option under the Cell category.

Note: You can also begin typing **nmos1v** in the *cell* name field of the Add Instance form. The auto-complete feature brings the *nmos1v* cell name into the field for easy selection.

As you make your selection, note that the Add Instance form modifies itself.

5. Select the **symbol** view with the left mouse button. Your Library Manager should be like what you see in the following snapshot.



The Add Instance form now displays some information along with auto-filled text fields, as shown in the following snapshot. Give first *nmos1v* as **M1** and another as **M2** in the **Names** field. Click on **Sideways** to get the mirror view of the instance:

The 'Add Instance' dialog box is shown with the following details:

- Library:** gpdk045
- Cell:** nmos1v
- View:** symbol
- Names:** M1
- Add Wire Stubs at:** ☒ all terminals, ☐ registered terminals only
- Array:** Rows: 1, Columns: 1
- Orientation:** Rotate, Sideways (selected), Upside Down
- Model Name:** g45n1svt
- Multiplier:** 1
- Length:** 45n M
- Total Width:** 120n M
- Finger Width:** 120n M
- Fingers:** 1
- Threshold:** 120n M
- Apply Threshold:** ☐
- Diff Cont:** ☒
- Gate Connection:** None
- Use DFM Rules:** Minimum
- S/D Metal Width:** 60n M
- Switch S/D:** ☐
- Bodytie Type:** None
- Edit Area & Perim:** ☐
- Drain diffusion area:** 16.8f
- Source diffusion area:** 16.8f
- Buttons:** Hide, Cancel, Defaults, Help

6. Close the **Library Browser – Add Instance** form. Leave the Add Instance form open with *Library*, *Cell* and *View* pull-down cyclic fields populated.

Adding Components to a Schematic

1. Click the **Hide** button toward the bottom of the form.

Follow the next steps diligently, as they will allow you to place the chosen *nmos* transistor on your schematic.

- a. Move your *cursor* into the schematic window. A highlighted symbol is displayed. It is ready to be placed on the schematic.
- b. **Left-click** to place the symbol on the schematic.

- c. Place the mirror view of *nmos1v* instance (i.e., **M1**) on the schematic. Also, place another instance of *nmos1v* with its normal view (i.e., **M2**), in such a way that both the gates of *nmos* should be facing each other.

Note: Observe that the device has been added to the schematic and the symbol has an instance name. A highlighted symbol of *nmos1v* remains and moves with the cursor.

- d. Click the <Esc> key to quit placement mode.

- 2. Press the **I** key.

The Add Instance form is displayed again, with the library, cell, and view fields filled. Likewise, a highlighted symbol tracks the cursor in the schematic window.

- a. Move the cursor into the Add Instance form. Change the library field to **analogLib** from the library column. Click in the cell text field and choose **idc**.

Note: You can use the **Browse** button to search for *idc*. However, entering *idc* directly into the text field saves time.

- b. Set the parameter as follows:

DC current: 100u A

- c. Your form should be similar to what you see in the following snapshot:

The 'Add Instance' dialog box is shown with the following settings:

- Library: analogLib
- Cell: idc
- View: symbol
- Names: (empty)
- Add Wire Stubs at: ☒ all terminals, ☐ registered terminals only
- Array: Rows: 1, Columns: 1
- Buttons: Rotate, Sideways, Upside Down
- Noise file name: (empty)
- Number of noise/freq pairs: 0
- DC current: 100u A
- AC magnitude: (empty)
- AC phase: (empty)
- Multiplier: (empty)
- Temperature coefficient 1: (empty)
- Temperature coefficient 2: (empty)
- Nominal temperature: (empty)
- XF magnitude: (empty)
- PAC magnitude: (empty)
- PAC phase: (empty)
- Buttons: Hide, Cancel, Defaults, Help

- d. **Hide** the form. Place this instance of *idc* on the schematic above the **MI** instance.

3. Press the **I** key.

Under the library, click in the cell text field, choose **vdc** in the Add Instance form, and change the parameter as follows:

DC voltage: vds

Hide the form. Place this instance of *vdc* on the schematic, on the right side of the *M2* instance.

Add Instance

Library: analogLib Browse

Cell: idc

View: symbol

Names:

☒ Add Wire Stubs at:
☐ all terminals ☒ registered terminals only ...

Array: Rows: 1 Columns: 1

Rotate Sideways Upside Down

Noise file name:

Number of noise/freq pairs: 0

DC current: 100u A

AC magnitude:

AC phase:

Multiplier:

Temperature coefficient 1:

Temperature coefficient 2:

Nominal temperature:

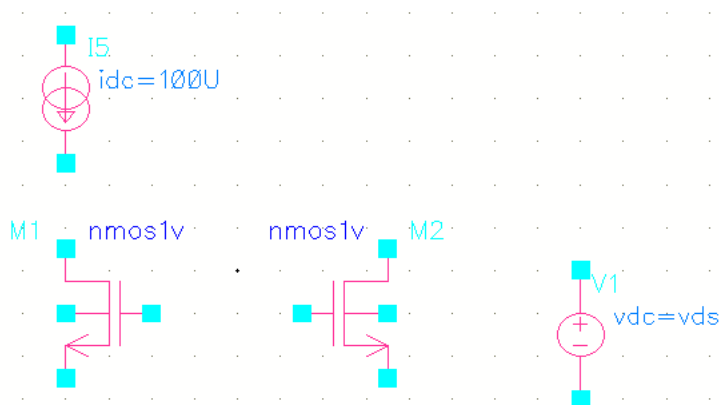
XF magnitude:

PAC magnitude:

PAC phase:

Hide Cancel Defaults Help

4. You can use the keyboard arrow keys to move up and down the schematic. Your canvas should be similar to what you see in the following snapshot.

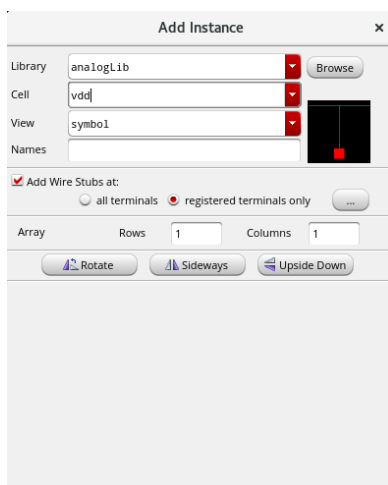


Adding Ground and VDD to the Schematic

1. Zoom out with the [key, so you have room to place the power and ground symbols.
2. Press **I** to display the Add Instance form
3. In the **Cell** text field of the Add Instance form, replace *vdc* with **gnd** under the same *analogLib* library.

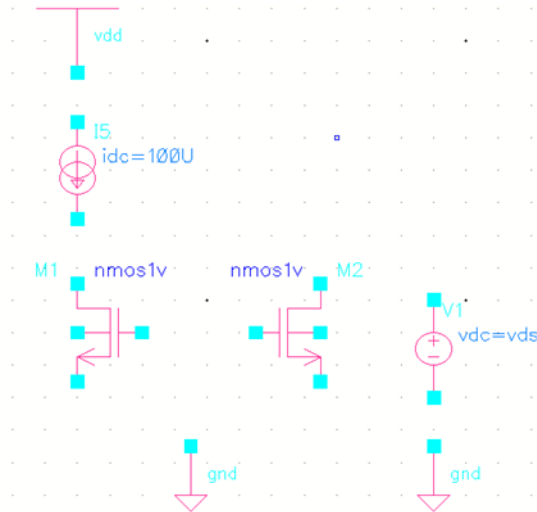


4. Move the cursor into the schematic and place a common *gnd* symbol below the sources of **M1** and **M2** devices. Also, place another *gnd* below the **vdc** instance.
5. In the **Cell** text field of the Add Instance form, replace *gnd* with **vdd** under the same *analogLib* library.



- Then press the **Esc** key. This will *close* the Add Instance form.

Important: Pressing **Esc** cancels any active command in the schematic window.



Creating Wires to a Schematic

Add wires to connect components in the design.

- Choose **Create – Wire (narrow)** from the pull-down menu. You can also use the **Create Narrow Wire** button from the Create toolbar section.

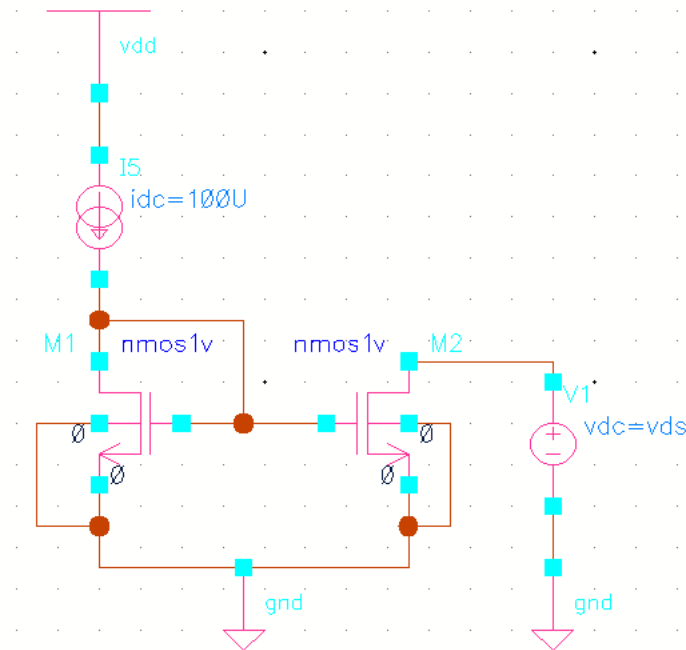
Note: Lowercase **w** is the bindkey for a *Narrow wire*, **Shift-W** for a *Wide wire*.

- In the schematic window, click one of your components' terminals, such as the *drain* terminal of **M1**, to make it the first point for your wiring.

Note: The wire router cannot always find a path between two points. If this happens, a dotted flight line is drawn between the points and establishes connectivity. If flight lines occur in your lab, delete them and click on the intermediate points to guide the router as your design is wired.

Follow the prompts at the bottom of the design window, move the mouse cursor, and **left-click** the destination point for your wire, such as nmos gate. A net is routed.

- Continue wiring the schematic, as shown in the following snapshot.



The exact topology shown in the reference schematic is not as important as the connectivity.

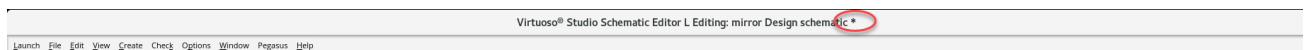
Caution: Be sure to connect the substrate for the *nmos1v* transistor to *gnd* respectively.

Important: If you make a mistake while wiring, select that section of the wire and press the **<Delete>** key to delete that part of the wire.

- Press **Esc** with your cursor in the schematic window to cancel wiring when finished wiring.

Saving a Design

Note on the title bar that there is an asterisk next to the name schematic. This indicates that your schematic has been edited and needs to be saved.

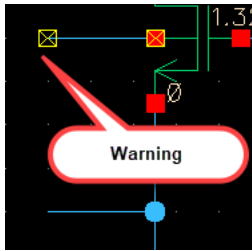


- Click the **Check and Save** icon in the schematic editor window.



- Or choose **File – Check and Save** or click **Shift + X**.

Note: A yellow box appears around all the unconnected pins and wires. If there are any, close the associated warning message and complete the connectivity.

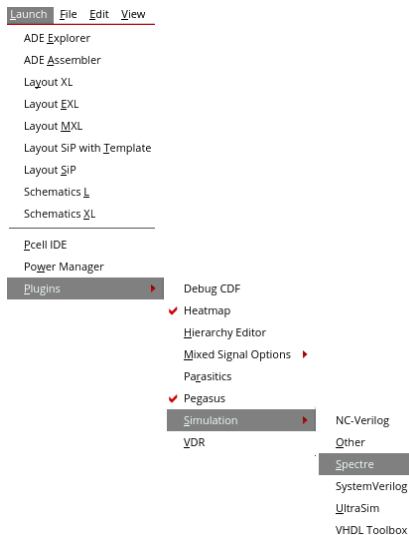


3. Check the CIW output area for any messages after a successful check and save.

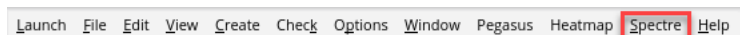
```
INFO (SCH-1170): Extracting "mir_sch schematic"
INFO (SCH-1426): Schematic check completed with no errors.
INFO (SCH-1181): "mirror mir_sch schematic" saved.
```

Netlisting the Design

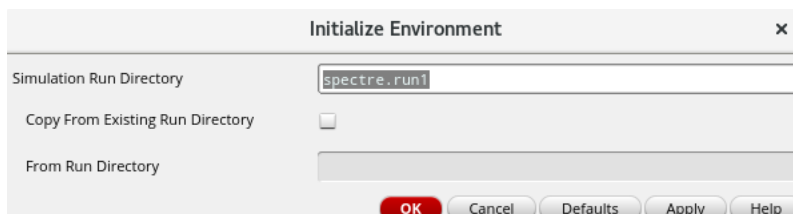
1. From the Schematic Editor window, choose **Launch – Plugins – Simulation – Spectre**.



Note: This attaches the Spectre Classic Simulator in the menu bar.



2. From the menu bar, choose **Spectre – Initialize**.



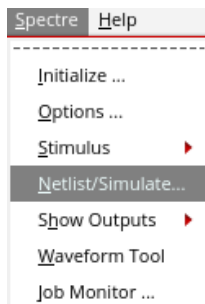
3. Click **OK** in the Initialize Environment form to create simulation result databases in the *spectre.run1* default simulation run directory.
 - a. Click **OK** in the next form that opens (shown in the following snapshot), verifying the design to be netlisted.



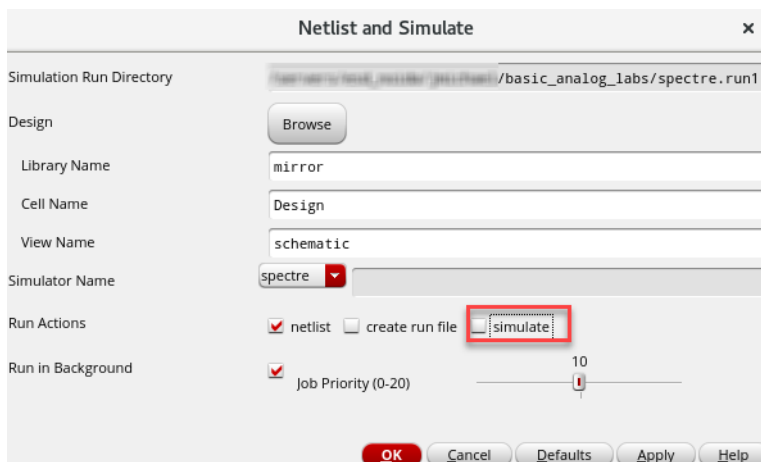
The 'Initialize Environment' dialog box is shown. It contains the following fields and controls:

- Simulation Run Directory:** /basic_analog_labs/spectre.run1
- Simulator Name:** spectre (dropdown menu)
- Design:** Browse button
- Library Name:** mirror
- Cell Name:** Design
- View Name:** schematic
- Buttons:** OK, Cancel, Defaults, Apply, Help

4. Choose **Spectre – Netlist/Simulate** to open the Netlist and Simulate form.



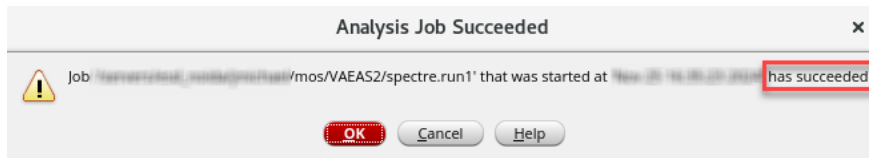
5. **Disable** the *Run Actions – simulate* option. Your form should look similar to the one you see in the following snapshot.



The 'Netlist and Simulate' dialog box is shown. It contains the following fields and controls:

- Simulation Run Directory:** /basic_analog_labs/spectre.run1
- Design:** Browse button
- Library Name:** mirror
- Cell Name:** Design
- View Name:** schematic
- Simulator Name:** spectre (dropdown menu)
- Run Actions:**
 - ☒ netlist
 - ☐ create run file
 - ☐ **simulate** (highlighted with a red box)
- Run in Background:**
 - ☒ Job Priority (0-20): 10
- Buttons:** OK, Cancel, Defaults, Apply, Help

6. Click **OK**. In the *Analysis Job Succeeded* form that comes up, click **OK**.



7. View the netlist file in the *spectre.run1* directory by choosing **Spectre – Stimulus – Edit Netlist File**.



Try to associate the connectivity in the netlist along with your drawn schematic.

8. **Close** the netlist window, leave the *Design* schematic view and CIW session running and proceed to the next lab.



Lab 1-2 Simulation Using ADE Explorer and Virtuoso VA

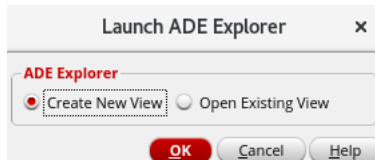
Objective: To simulate the current mirror circuit using ADE Explorer and Virtuoso VA.

In this lab, you

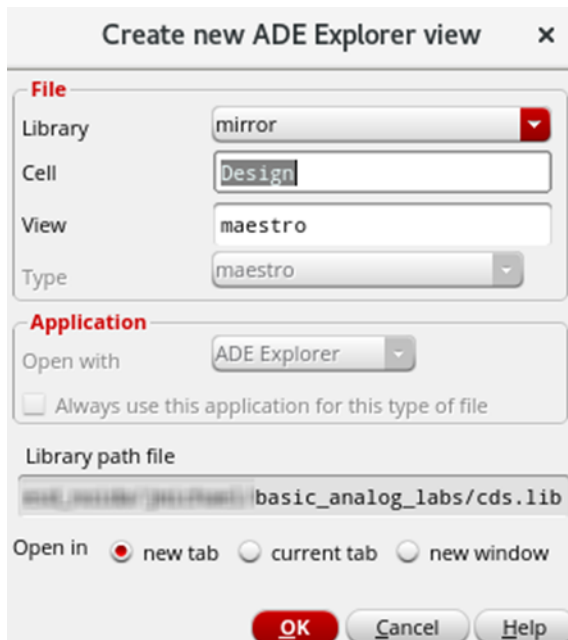
- ◆ Learn how to simulate the current mirror created in the previous lab by creating a new *maestro* view using Virtuoso ADE Explorer.

Starting the Virtuoso ADE Explorer and Creating a *maestro* View

1. Choose **Launch – ADE Explorer** from the *Design* schematic window to open the Launch ADE Explorer form.

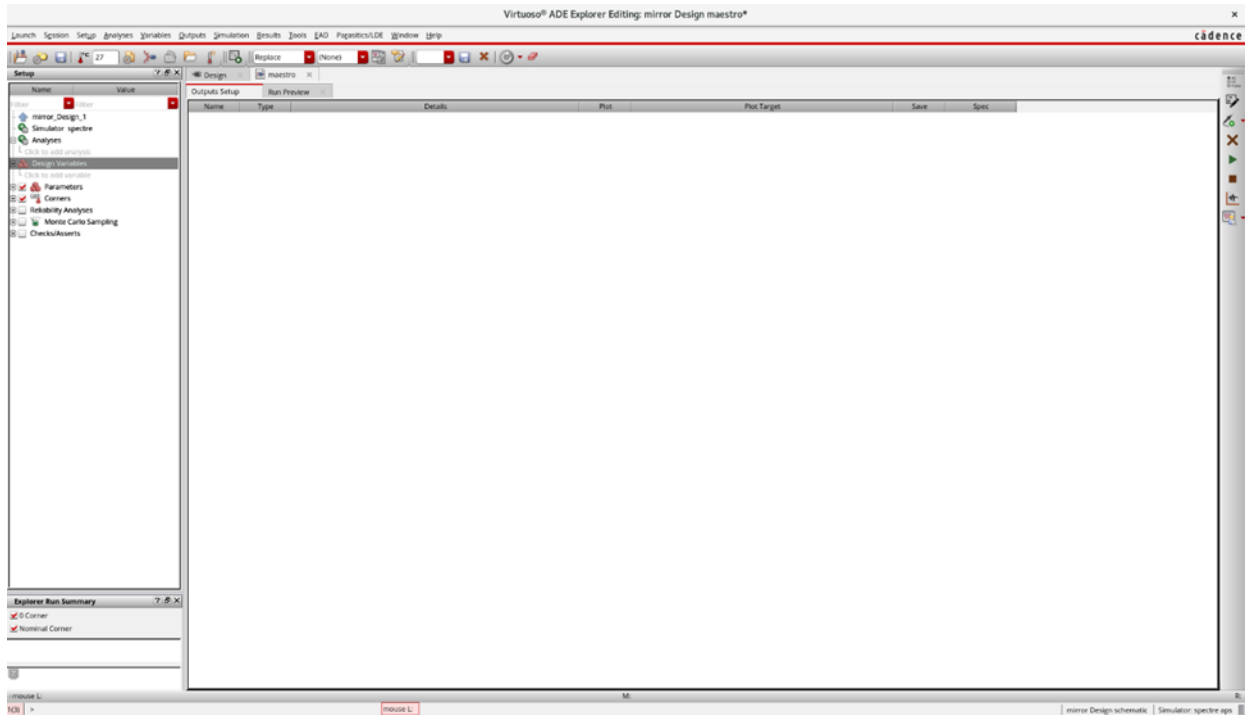


2. Select **Create New View** to open the **Create new ADE Explorer view** form.



3. The **maestro** view type is chosen by default, and so is the *Open with* the application, which is set to the **ADE Explorer**.

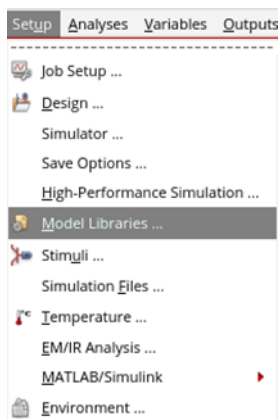
Leave the View name set to the default name, **maestro**. Choose to open in a new tab and click **OK** on this form to open the ADE Explorer environment in a new **maestro** tab.



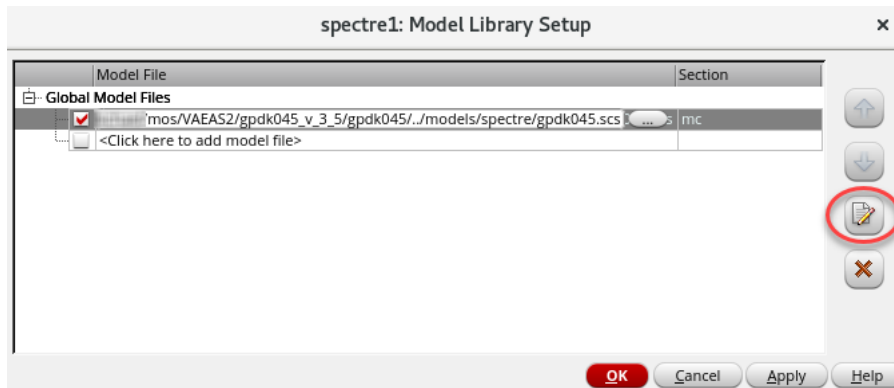
Setting the Model Libraries

The model library file contains the model files that describe the *nmos1v* device during simulation.

1. Choose **Setup – Model Libraries**.



The Model Library Setup form appears. You can see that there will be a *gpdk045 model library* added automatically. This is because, we have attached the **mirror** library to *gpdk045* in the beginning.



- a. To view the model file, select the model file. Click the **Edit/View Selected Files** button.



Close the model file after viewing.

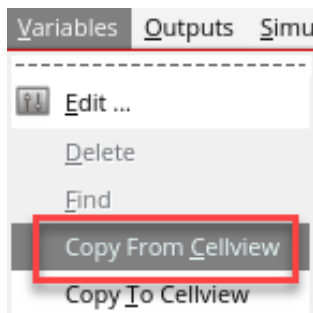
With the Model Library Setup form, you can include multiple models.

- b. On the Model Library Setup form, click **OK**.

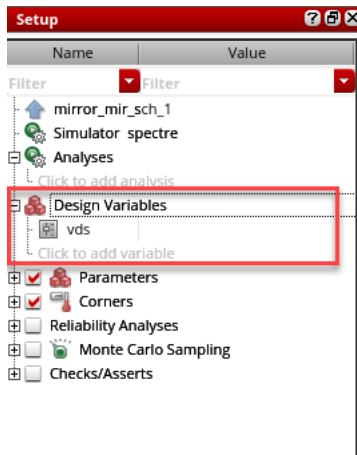
Setting Design Variables

The values of any design variables in the circuit must be set before running a simulation. If values have been set in the past, they appear in the Value column of the Design Variables section of the Setup assistant in the ADE Explorer form.

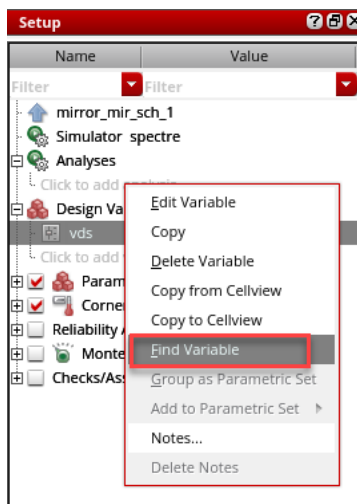
Note: If there are variables in the text files down in the hierarchy, retrieve them. Choose **Variables – Copy from Cellview** on the ADE Explorer menu.



The *vds* variables appear in a table in the Design Variables section.

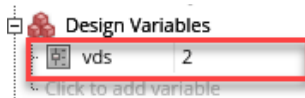


1. To locate the *vds* variable in the *current mirror* design, **right-click** the *vds* name and select **Find Variable**



The Find Variable operation takes you to the *current mirror* schematic tab and highlights the *vds* instance.

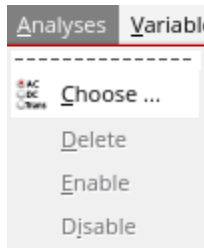
2. Move to the **maestro** tab and set the value of the *vds* variable to **2** by clicking the *Value* field after the variable *vds* and entering the new value. Press **Enter** to accept the change.



Choosing Analyses

This part demonstrates how to select **tran analyses** to complete your circuit during the simulation. You can select and run dc analysis on the *current mirror* design.

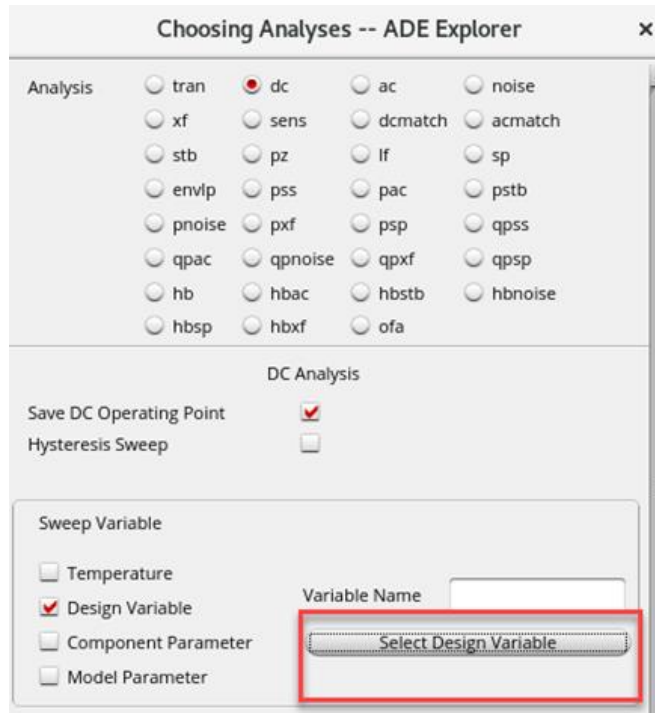
1. In the Simulation window, choose **Analyses – Choose**



2. Or click the **Choose Analysis** icon.



- a. The Choosing Analyses form appears.
 - b. Take a few moments to become familiar with this form because it is used often throughout the lab activities. It is also a dynamic form; the layout changes based on the selected analyses.
3. Set up a DC analysis.
 - a. In the Analysis section, select **dc**.
 - b. In the DC analysis section, select the **Save DC Operating Point** checkbox to turn it on.
 - c. In the **Sweep Variable** section, click on **Design Variable** to select the variable to be in the input or X-axis.



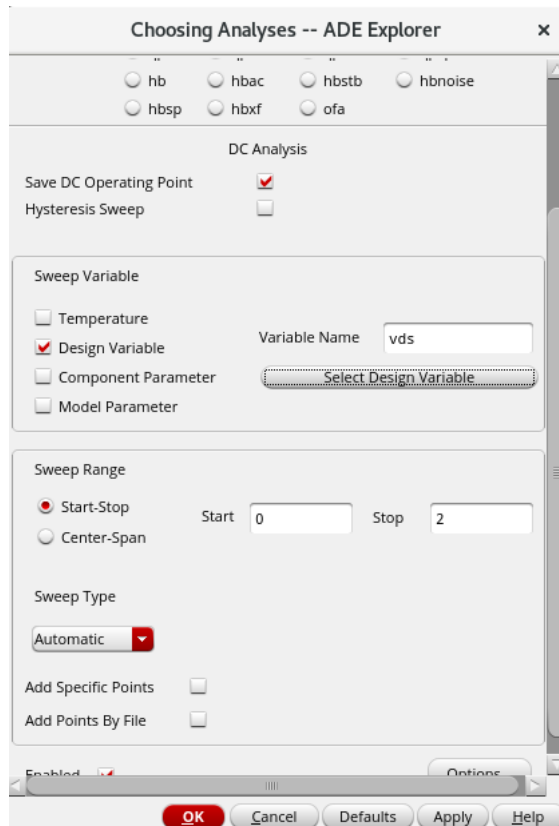
- d. Click on the **Select Design Variable** button, and in the form opened, click on the *vds* variable and click **OK**.



- e. In the **Sweep Range** section, click on the *Start-Stop* radio button and give the values as follows:

Start: 0, Stop: 2

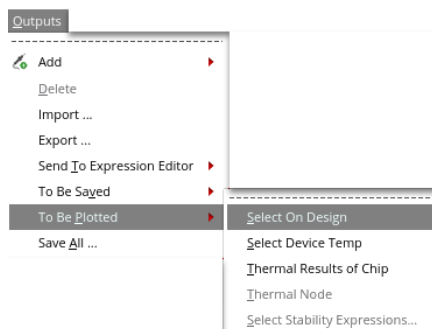
Ensure that the **Enabled** checkbox is selected. Your form should look similar to the one you see in the following snapshot. Click on **OK**.



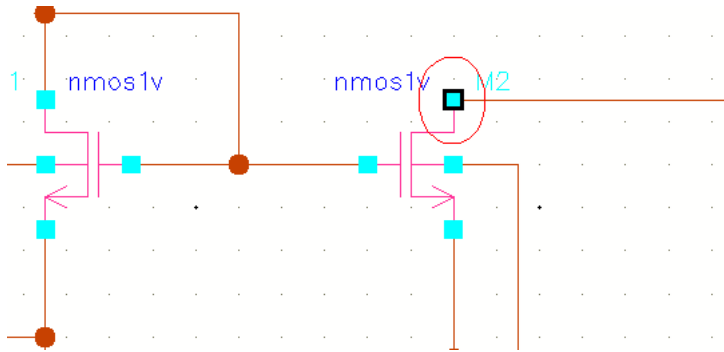
Saving and Plotting Simulation Outputs

The simulation environment is configured to save all public node voltages by default. It also saves the currents you select for plotting and saving.

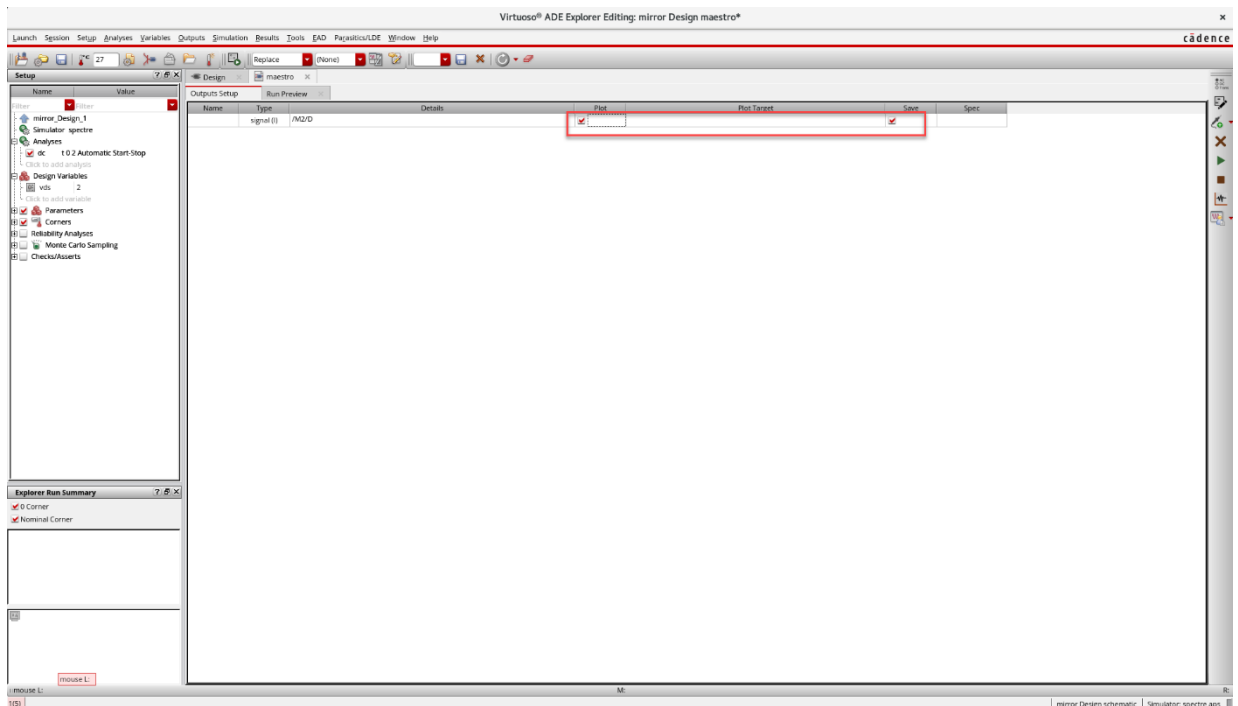
1. To select specific signals for saving and plotting, choose **Outputs – To Be Plotted – Select On Design** in the ADE Explorer window.



2. This brings you to the front of the schematic window for your selections.
 - a. Select the *drain* terminal for plotting. After selections, the terminal gets highlighted.



- b. Press **Esc** with your cursor in the schematic window to stop the selection process.
- c. Move to the **maestro** tab. Enable the plot and verify whether it looks like the one in the following snapshot:



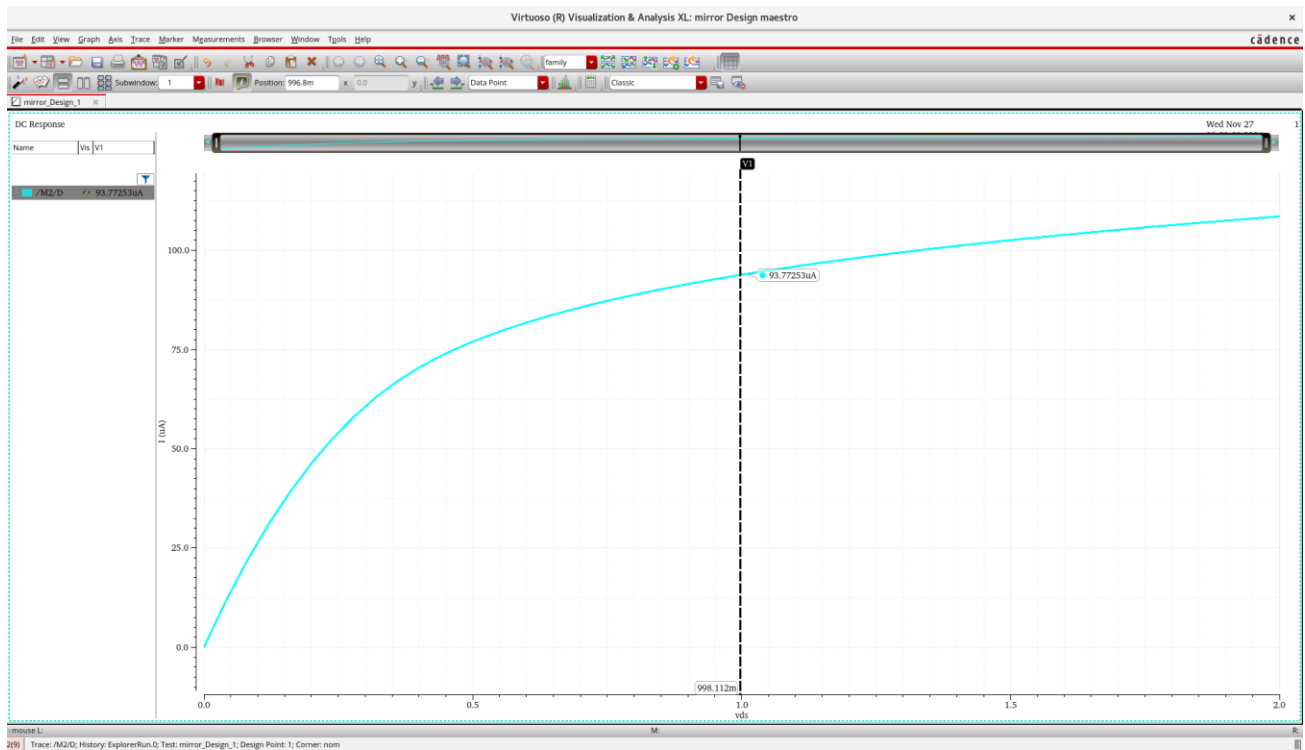
Running the Simulation

1. Choose **Simulation – Netlist and Run** or click the **Run Simulation** button  to recreate the netlist and run the simulation.
 - a. As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.
 - b. A log file shows the progress of the simulation. This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.

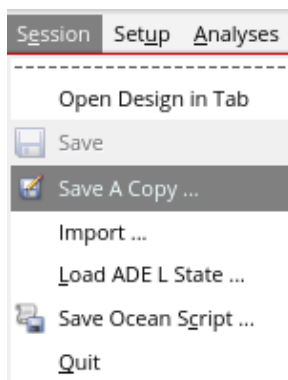
- c. The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

Tip: If you close this log file, you can open it again by going to Simulation – Output Log.

2. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso Visualization & Analysis XL window.



3. Click on the **Save** icon to save the graph. Move to the Maestro window and save the simulation settings by clicking on the **Session – Save a Copy**.

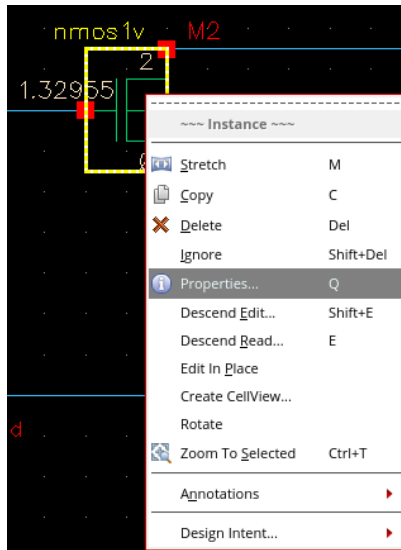


4. In the **Save a Copy** form, name the view and save it.



DC Parametric Analysis

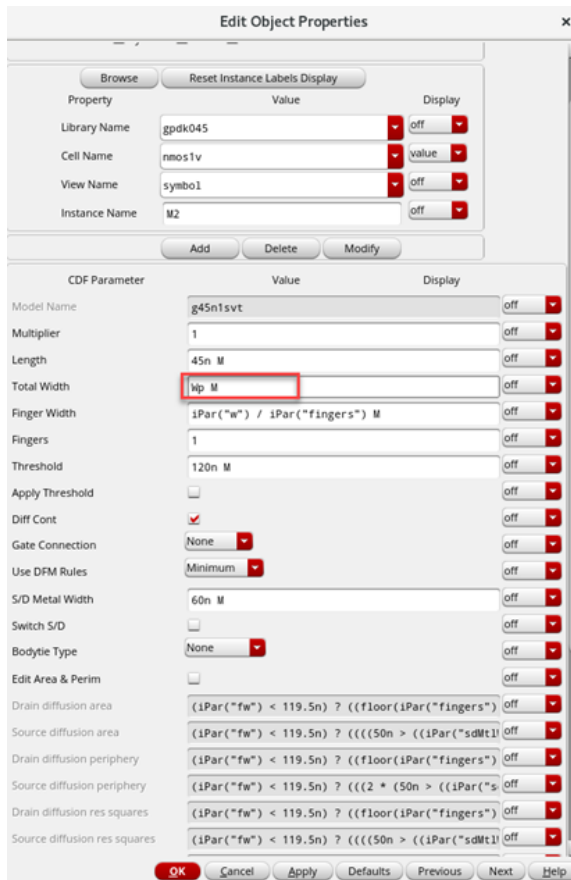
1. For Parametric Analysis, move to the *Design* schematic view. Select the **M2** instance and select **Properties** in the drop-down menu by **right-clicking** on the **M2**.



2. On the **Edit Object Properties** form, type in the **Total Width** field as follows:

Total Width: Wp

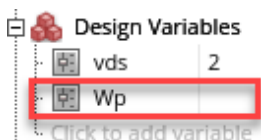
3. Your canvas should be similar to what you see in the following snapshot:



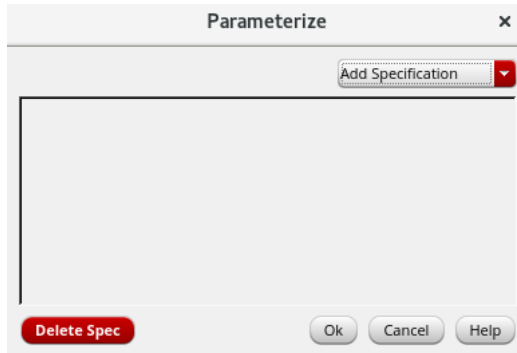
4. Choose **File – Check** and **Save** or **click Shift + X**.
5. Verify the CIW for warnings and errors.

Simulating the Parametric Analysis

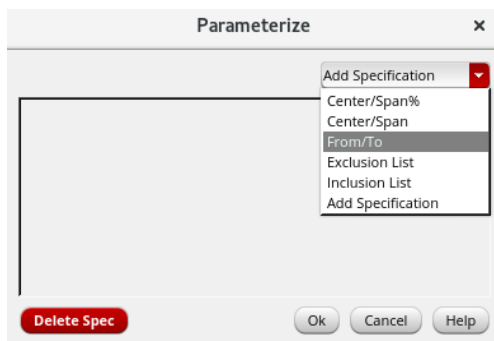
1. Choose **Launch – ADE Explorer** from the *Design* schematic window to open the Launch ADE Explorer form.
2. Move to the **Maestro** tab with the already filled settings of the previous analysis. Choose **Variables – Copy from Cellview** on the ADE Explorer menu. This includes the **Wp** variable in the **Design Variables** section.



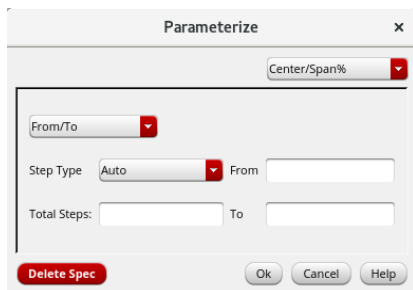
- For the W_p click on the value field to enable the button  of the **Parameterize** form. Click on it to open the form once it is enabled.



- In the drop-down of **Add Specification**, click on **From/To** to add the *Spec*.



- In the **Step Type** drop-down menu, click on **LinearStepCount**.



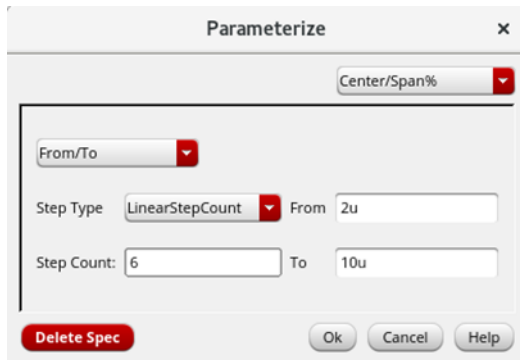
- Type the values as follows:

From: 2u

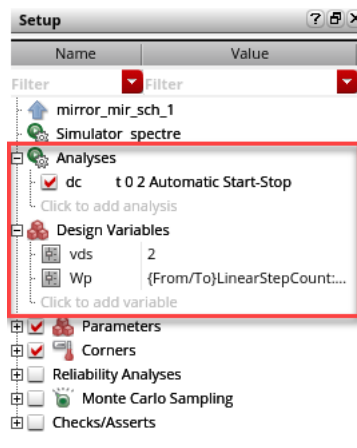
To: 10u

Step Size: 6

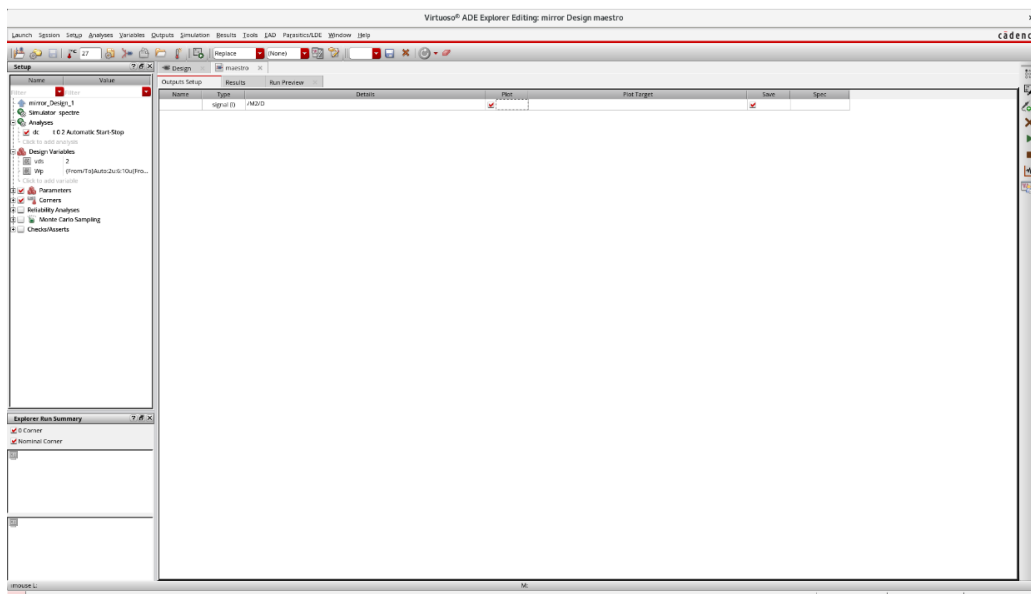
Your form should look similar to the one you see in the following snapshot. Click on **OK**.



- Keep the settings of the Analysis form as filled for the previous analysis. Your canvas should be similar to what you see in the following snapshot:



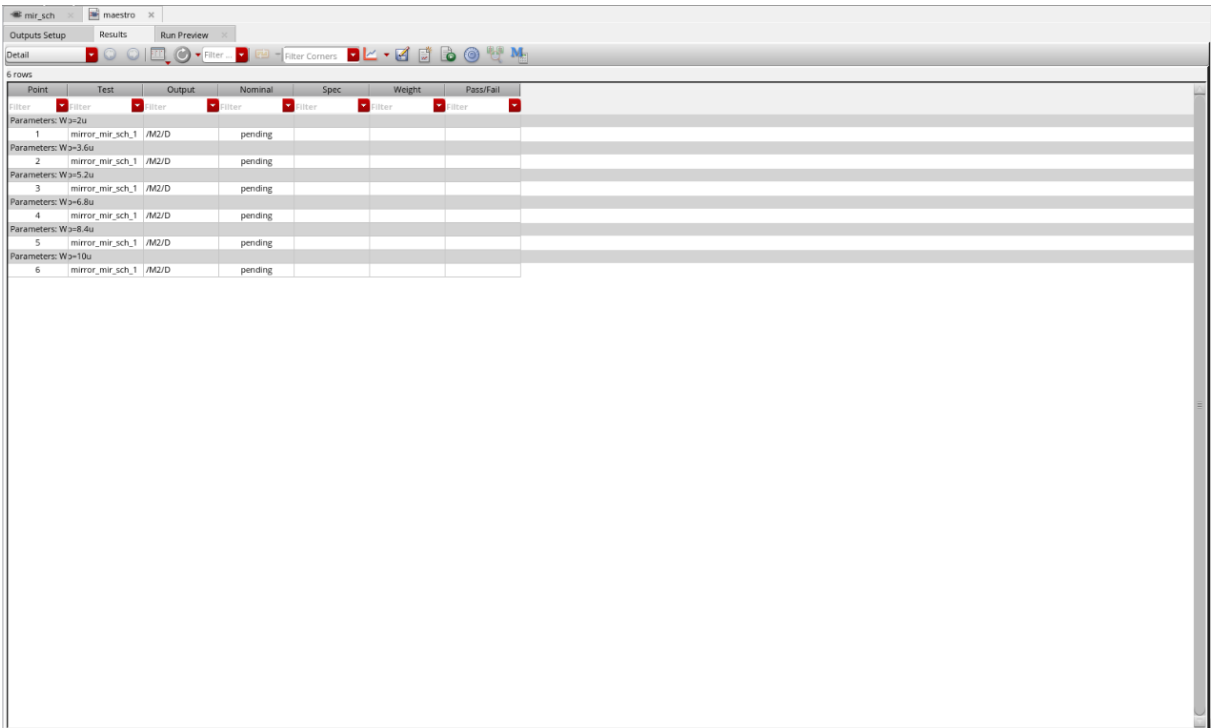
- Keep the settings in the **Outputs Setup** in the **Maestro** tab, as filled for previous analysis. Your canvas should be similar to what you see in the following snapshot:



9. Choose **Simulation – Netlist and Run** or click the **Run Simulation** button  to recreate the netlist and run the simulation.
- a. As an alternative to the previous step, you can choose **Simulation – Run** from the menu if you have already created your netlist.
 - b. A log file shows the progress of the simulation for parametric analyses of W_p . This is called the *spectre.out* file and is stored in a hierarchical psf directory inside your *./simulation* directory.
 - c. The log file also lists warnings and errors if there are problems in the input data, which often is missing model files.

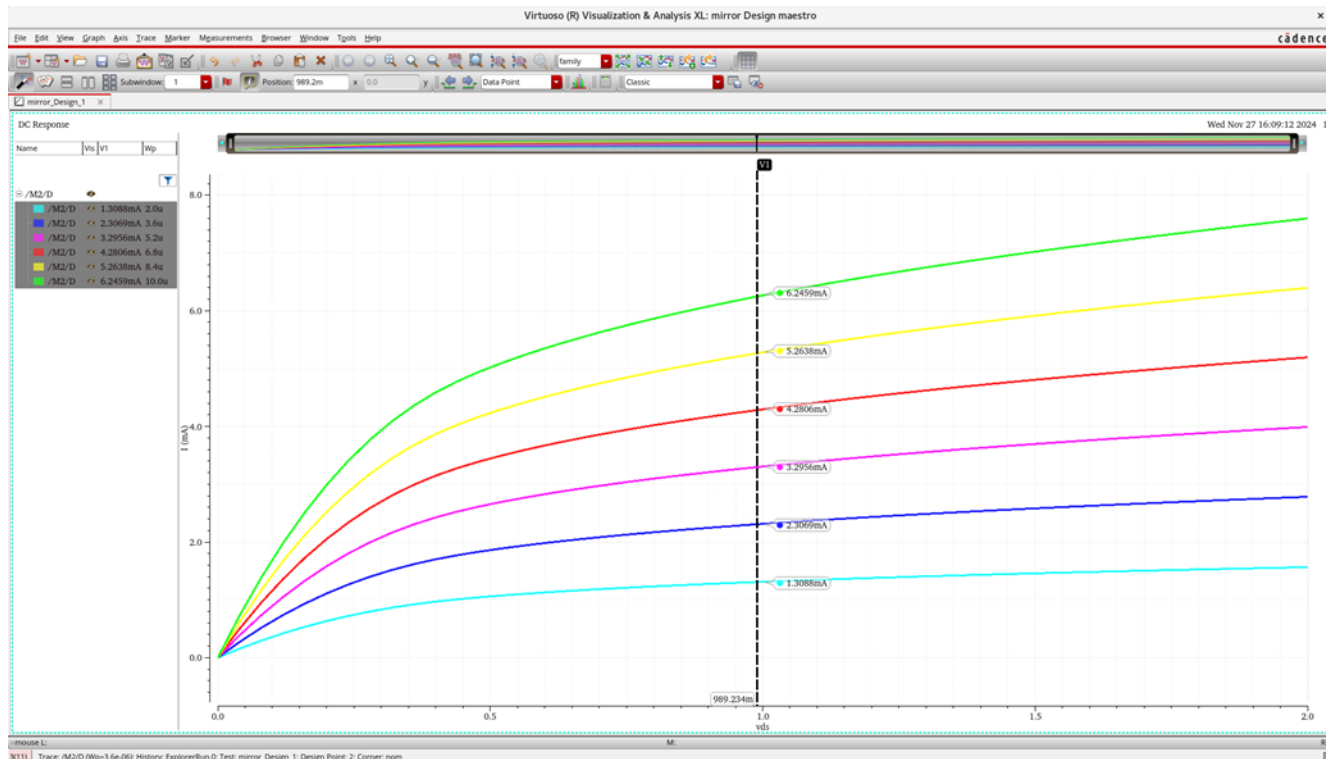
Tip: If you close this log file, you can open it again by going to the Simulation – Output Log.

10. The Results tab of the Maestro shows the status of each trace or value of W_p according to the *Step Size* given in the **Parameterize** form.

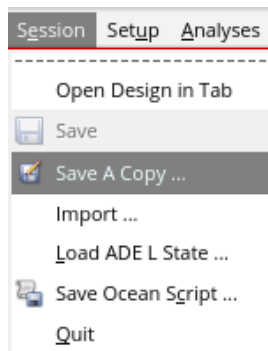


Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
1	mirror_mir_sch_1	/M2/D	pending			
2	mirror_mir_sch_1	/M2/D	pending			
3	mirror_mir_sch_1	/M2/D	pending			
4	mirror_mir_sch_1	/M2/D	pending			
5	mirror_mir_sch_1	/M2/D	pending			
6	mirror_mir_sch_1	/M2/D	pending			

11. When the simulation finishes, the DC Response plot is plotted automatically in the Virtuoso Visualization & Analysis XL window.



12. Click on the **Save** icon to save the graph. Move to the Maestro window and save the simulation settings by clicking on the **Session – Save a Copy**.



13. In the **Save a Copy** form, name the view and save it.

